

NATHAN DOUGLAS

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Platform engineer with 14+ years building distributed infrastructure and observability-heavy systems. Focused on making complex systems behave predictably: migration without downtime, deployment without failure, correctness by design.

Exploring formal verification (TLA+) for distributed and ML-serving systems.

🔒 U.S. Public Trust (Tier 2, active) · Prior Top Secret clearance

EXPERIENCE

Agile Six Applications, Inc (contract to U.S. Department of Veterans Affairs)

Oct. 2020 – Present

Senior Platform Engineer

Feb. 2024 – Present

- Migrated VA.gov Check-In infrastructure between AWS accounts with zero downtime; converted CloudFormation to Terraform with automated testing to resolve configuration drift
- Engineered cross-product system tests for VA.gov, automating integrations with VistA EHR, Oracle Health, and VEText to verify PHI/PII compliance across critical veteran healthcare workflows

Technical Lead

Jul. 2023 – Feb. 2024

- Led Drupal 8→9 and 9→10 upgrades as Technical Lead for a 12-person CMS team, coordinating tenant teams and stakeholders with zero unplanned downtime
- Reduced CI runtime by 60% through test parallelization; architected event-driven content release pipeline serving VA.gov's static content with zero deployment-related failures
- Maintained Authority to Operate (ATO) for VistA-integrated PHI/PII systems, resolving regulatory identity gaps through audits and tabletop exercises
- Recruited senior engineers and a QA architect to remediate systemic technical debt and flaky tests

Senior Software/DevOps Engineer

Oct. 2020 – Jul. 2023

- Spearheaded QA initiative to broaden unit test coverage, refine behavioral and accessibility tests, and mentor engineers on test and system design
- Implemented a service to monitor and respond to corner cases in GitHub Actions self-hosted runner registration and orchestration, improving CI reliability

Convention News Company

Jan. 2016 – Oct. 2020

Senior Software/DevOps Engineer

- Led migration from Acquia to AWS, reducing annual infrastructure costs by 70% and eliminating legacy stability issues
- Engineered a headless Drupal and Next.js architecture on AWS, integrating Stripe and AWS Cognito for subscription services with independent scaling
- Built Lambda-based subscriber services and data pipelines with CloudWatch and Slack for real-time monitoring; optimized performance via Varnish and Redis caching
- Established CI/CD pipelines and containerized development environments, increasing deployment frequency to multiple times daily
- Mentored junior engineers on clean code practices and peer review, improving ticket turnaround and codebase maintainability

DEVONtechnologies

Jan. 2012 – Dec. 2015

Software Engineer

- Developed client-side features and synchronization functionality for DEVONthink across macOS and iOS, working across cloud storage and peer-to-peer sync backends
- Took ownership of a complex sync subsystem during a high-risk engineer transition, supporting release stabilization, defect resolution, and customer issue triage
- Contributed to release engineering and iterative debugging in a legacy Objective-C codebase with significant performance, reliability, and memory-management constraints

TECHNICAL SKILLS

Languages: Rust, Python, Bash, JavaScript, TypeScript, PHP, Go, C, Objective-C, Clojure, $\LaTeX$

Infrastructure-as-Code: Terraform, Ansible, CloudFormation, Terragrunt

Orchestration: Kubernetes, Docker, Nomad

CI/CD: GitHub Actions, Jenkins, CodeBuild/CodePipeline

Observability: Prometheus, Grafana, OpenTelemetry, Datadog, CloudWatch

Networking/Security: Consul, Envoy, Vault, Cilium

PROJECTS

-  **Goldentooth:** Personal research lab: multi-node Kubernetes cluster exploring infrastructure verification (TLA+), GPU compute, full observability, and GitOps deployment (*Talos, Kubernetes, Flux CD, TLA+*)
-  **Bitwit:** Spaced repetition platform for CS education with time-based grading and parameterized card generation (*TypeScript, AWS Lambda, DynamoDB*)
-  **Whispers:** Gossip-based mesh network with fraud-resistant distributed optimization via audited CRDTs (*Rust*)
-  **Distillation:** Knowledge distillation with Jacobian/Hessian regularization and adversarial training (*Python, PyTorch*)
-  **Blackpool:** Stack-based bytecode VM with mark-and-sweep GC, closures, and inheritance (*Rust*)
-  **Longtable:** Rule-based simulation engine with LISP DSL, immutable world state, and time-travel debugging (*Rust*)
-  **Greenstone:** Cycle-accurate NES emulator with precise timing and hardware modeling (*Rust*)
-  **Clork:** Zork I port with an OpenAI Gym-style RL training API (*Clojure*)

Open-source contributor to Terraform, OpenTofu, Argo CD, Datadog, Couchbase, and Drupal, among others

EDUCATION

University of Nevada, Las Vegas
Bachelor of Arts in Computer Science

Las Vegas, NV
Dec. 2011

LANGUAGES

English: Native
French: A2/B1 (in progress, targeting C1 by 2028)